

Phase I/II Operating Characteristics

Cycle 2 shared additive versus updated cycle 2 and cycle 3 multicycle additive

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.3	0.3	0.4	0.5	0.5	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
OBD selection (%) - one-stage rows, then two-stage rows						
C2 shared additive one-stage	9.1	5.6	2.6	0.9	0.4	81.4
C2 multicycle one-stage	8.5	5.7	2.9	1.7	0.2	81
C3 multicycle one-stage	9.2	7.5	3.3	1.4	0.1	78.5
C2 shared additive two-stage	8.7	6.2	2.8	0.7	0.6	81
C2 multicycle two-stage	8.8	6	2.8	1.5	0.2	80.7
C3 multicycle two-stage	10.5	6.6	3.9	1.2	0.2	77.6
Mean administrations by dose - one-stage rows, then two-stage rows						
C2 shared additive one-stage	11	4.3	2.4	1	0.3	
C2 multicycle one-stage	10.6	4.8	2.8	1.2	0.4	
C3 multicycle one-stage	10.8	4.7	2.6	1.1	0.4	
C2 shared additive two-stage	10.9	4.4	2.5	1	0.3	
C2 multicycle two-stage	10.6	4.8	2.8	1.2	0.4	
C3 multicycle two-stage	10.8	4.7	2.7	1.2	0.4	
Scenario 16						
DLT rate	0	0.1	0.1	0.1	0.3	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
OBD selection (%) - one-stage rows, then two-stage rows						
C2 shared additive one-stage	23.3	21	19.8	21.2	13.7	1
C2 multicycle one-stage	23.5	22.4	19.8	18	14.8	1.5
C3 multicycle one-stage	24.3	21.6	18.8	19.5	14.7	1.1
C2 shared additive two-stage	25.9	21.1	17.5	20	14.6	0.9
C2 multicycle two-stage	22.3	22.5	18.6	19.6	15.6	1.4
C3 multicycle two-stage	23.7	22	17.6	19.8	15.8	1.1
Mean administrations by dose - one-stage rows, then two-stage rows						
C2 shared additive one-stage	7.3	6.6	5.7	5.7	4.6	
C2 multicycle one-stage	7.1	6.6	5.9	5.5	4.9	
C3 multicycle one-stage	7	6.4	5.7	5.9	4.9	
C2 shared additive two-stage	7.1	6.6	5.7	5.8	4.9	
C2 multicycle two-stage	6.9	6.4	6	5.7	5	
C3 multicycle two-stage	6.9	6.4	5.7	6	5.1	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 20						
DLT rate	0.1	0.1	0.3	0.4	0.5	
Efficacy rate	0.1	0.2	0.2	0.2	0.2	
OBD selection (%) - one-stage rows, then two-stage rows						
C2 shared additive one-stage	4.8	45.6	29.2	13.5	2	4.9
C2 multicycle one-stage	4.6	44.1	29.6	13.4	3	5.3
C3 multicycle one-stage	3.6	46.5	28.8	13.1	3.2	4.8
C2 shared additive two-stage	5.8	46.7	29.7	11.2	2.1	4.5
C2 multicycle two-stage	5.1	45	30.1	11.5	2.9	5.4
C3 multicycle two-stage	4.7	47.4	27.9	12.6	1.8	5.6
Mean administrations by dose - one-stage rows, then two-stage rows						
C2 shared additive one-stage	5.2	11	7.7	4.2	1.4	
C2 multicycle one-stage	5.2	10.7	7.8	4.3	1.7	
C3 multicycle one-stage	5.2	10.6	7.8	4.3	1.7	
C2 shared additive two-stage	5	10.3	7.9	4.7	1.6	
C2 multicycle two-stage	5	10.2	8	4.7	1.9	
C3 multicycle two-stage	5.1	10.2	8	4.5	1.9	
Scenario 24						
DLT rate	0	0	0.1	0.1	0.2	
Efficacy rate	0.1	0.1	0.2	0.5	0.5	
OBD selection (%) - one-stage rows, then two-stage rows						
C2 shared additive one-stage	1	1.1	4.9	59.4	33.6	0
C2 multicycle one-stage	0.9	1	5.1	60.2	32.7	0.1
C3 multicycle one-stage	0.6	1.7	6.3	58.1	33.2	0.1
C2 shared additive two-stage	1.2	1.4	5.7	56.6	35.1	0
C2 multicycle two-stage	1	0.9	5.7	55.4	36.9	0.1
C3 multicycle two-stage	0.2	1.9	5.2	59.6	33.1	0
Mean administrations by dose - one-stage rows, then two-stage rows						
C2 shared additive one-stage	3.4	3.5	4.4	12.2	6.5	
C2 multicycle one-stage	3.5	3.4	4.5	12.2	6.4	
C3 multicycle one-stage	3.4	3.5	4.5	12	6.5	
C2 shared additive two-stage	3.4	3.4	4.2	10.7	8.2	
C2 multicycle two-stage	3.4	3.4	4.5	10.6	8	
C3 multicycle two-stage	3.4	3.4	4.3	11.2	7.7	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.1	0.2	0.3	0.5	0.6	
Efficacy rate	0.1	0.2	0.4	0.4	0.5	
OBD selection (%) - one-stage rows, then two-stage rows						
C2 shared additive one-stage	10	22.9	46.3	11.1	1	8.7
C2 multicycle one-stage	11.3	21.9	48.3	11.7	0.5	6.3
C3 multicycle one-stage	10.1	22.8	48.1	11.4	1.2	6.4
C2 shared additive two-stage	10.5	22.3	47.2	10.9	1.2	7.9
C2 multicycle two-stage	10.7	23.6	49.3	9.3	1	6.1
C3 multicycle two-stage	9.7	21.9	50.8	9.9	1.1	6.6
Mean administrations by dose - one-stage rows, then two-stage rows						
C2 shared additive one-stage	7.1	7.8	10	3.6	0.7	
C2 multicycle one-stage	7	7.8	10.3	3.5	0.6	
C3 multicycle one-stage	7	7.8	10.1	3.8	0.6	
C2 shared additive two-stage	6.8	7.5	9.7	4.1	1.1	
C2 multicycle two-stage	6.8	7.7	9.9	3.9	1	
C3 multicycle two-stage	6.7	7.6	9.7	4.2	1	
Scenario 38						
DLT rate	0.1	0.1	0.1	0.1	0.3	
Efficacy rate	0.1	0.1	0.1	0.1	0.5	
OBD selection (%) - one-stage rows, then two-stage rows						
C2 shared additive one-stage	4.2	6	7.8	11.5	66.2	4.3
C2 multicycle one-stage	4.8	5.6	9.5	11.9	64.2	4
C3 multicycle one-stage	4.2	5.5	8.2	12.3	66.2	3.6
C2 shared additive two-stage	4.6	6.8	7.7	12	64.6	4.3
C2 multicycle two-stage	5.2	5.3	10.1	11	64.1	4.3
C3 multicycle two-stage	4.9	5.5	8.5	12.5	65	3.6
Mean administrations by dose - one-stage rows, then two-stage rows						
C2 shared additive one-stage	4.5	4.4	4.9	5.4	10.6	
C2 multicycle one-stage	4.7	4.3	5.2	5.4	10.2	
C3 multicycle one-stage	4.7	4.4	4.9	5.3	10.6	
C2 shared additive two-stage	4.4	4.4	4.9	5.5	10.7	
C2 multicycle two-stage	4.7	4.3	5.2	5.4	10.3	
C3 multicycle two-stage	4.6	4.3	4.9	5.4	10.7	

Mean Unique Patients by Method

Mean number of unique patients per simulated trial

Scenario	1	16	20	24	27	38	Mean
C2 shared additive one-stage	12.3	18.2	18.9	19.6	19.4	18.9	17.9
C2 multicycle one-stage	12.7	18.2	18.9	19.5	19.5	18.9	18
C3 multicycle one-stage	10.6	14.4	15.7	16.5	16.5	15.5	14.9
C2 shared additive two-stage	12.4	18.3	19	19.7	19.6	18.9	18
C2 multicycle two-stage	12.7	18.2	19.1	19.6	19.6	18.9	18
C3 multicycle two-stage	10.7	14.5	15.9	16.7	16.7	15.6	15

Values are reported from Total_Unique_Patients and shown to one decimal. The Mean column is the unweighted mean across the six scenarios.

Rows are grouped by allocation. Shared C2 is continuous; multicycle C2 and C3 source rows are IPDE-first enrollment.